

Development of a PPG-based Low-cost Cuffless Continuous Blood Pressure Monitoring with Vital Sign Detection System

Abstract—The paper discusses designing cuffless continuous blood pressure monitoring system using non-invasive sensing method. The proposed low-cost solution, with an estimated price of around 50 USD, integrates a MAX32664D biometric sensor hub with a MAX30102 PPG sensor to enable real-time monitoring of systolic and diastolic blood pressure, mean arterial pressure (MAP), heart rate, oxygen saturation (SpO₂), and sensor-based skin temperature measurement. The system employs calibration-based trending algorithms that analyze PPG signal morphology changes to estimate blood pressure variations relative to a baseline reference. Clinical validation was conducted on 20 participants (10 young adults aged 20-28 and 10 elderly participants aged 45-64) with measurements compared against traditional cuff-based sphygmomanometers and pulse oximeters. Results demonstrate high accuracy rates: 96.53% for systolic BP, 94.99% for diastolic BP, and 96.53% for MAP in young adults; and 95.29% for systolic BP, 94.91% for diastolic BP, and 96.01% for MAP in elderly participants. The portable, battery-powered device addresses daily healthcare needs by eliminating patient discomfort associated with inflatable cuffs while enabling continuous monitoring essential for intensive care applications.

Keywords—blood pressure trend (BPT), photoplethysmogram (PPG), sphygmomanometer, blood pressure, systolic, diastolic, biometric hub sensor, cuffless continuous monitoring

I. INTRODUCTION

Hypertension (HTN) is an increasingly important medical and public health problem. In Bangladesh, approximately 20% of adult and 40–65% of elderly people suffer from HTN. High incidence of metabolic syndrome, and lifestyle-related factors like obesity, high salt intake, and less physical activity may play important role in the pathophysiology of HTN [1]. The COVID-19 pandemic highlighted the need for remote, contact-minimizing monitoring solutions to reduce infection risk and healthcare workload. Continuous blood pressure (BP) monitoring is essential in critical care units (CCUs) and intensive care units (ICUs) to detect hemodynamic instability in real time." It allows for timely intervention for both hypertension and hypotension to preserve appropriate organ perfusion and prevent complications. Regular or ongoing blood pressure (BP) monitoring is essential for the early identification of hypertensive episodes in preeclampsia. One of the main markers of the course of the disease, elevated blood pressure is linked to risks like fetal growth restriction, placental insufficiency, stroke, and eclampsia. By enabling timely antihypertensive medication intervention, timely blood pressure monitoring lowers maternal and fetal morbidity and mortality [2]. Numerous physiological and practical factors make it difficult to measure blood pressure (BP) accurately in both pediatric and elderly populations. Reduced vascular compliance in older adults due to arterial stiffening can result in systematic overestimation when using oscillometric devices based on cuffs. Measurement accuracy is further weakened by orthostatic blood pressure changes, irregular heart rhythms (such as atrial fibrillation), and challenges with correctly

positioning the cuff. One of the main causes of errors in pediatric patients is improper cuff sizing; readings that are too big or too small in relation to arm circumference are erroneous. Furthermore, accurate blood pressure measurement is made more difficult by patient movement, anxiety (white coat effect), and the scarcity of validated pediatric devices. Clinical decision-making is weakened by these problems, which also lead to measurement bias and increased variance [3]. Continuous blood pressure (BP) can be measured invasively by inserting a catheter into an artery, but this method is complex, carries infection risks, and may damage vessels [4]. Non-invasive cuff-based sphygmomanometers remain the clinical standard; however, they are uncomfortable, provide only intermittent readings, and are unsuitable for tracking BP variations throughout the day [5]. Advances in non-invasive sensing have led to cuffless approaches, offering continuous, discreet, and user-friendly monitoring. One such method is Blood Pressure Trending (BPT), which estimates BP by analyzing photoplethysmography (PPG) signal morphology relative to a calibrated baseline. Unlike absolute BP measurement, BPT provides reliable trends rather than exact values, enabling long-term monitoring without repeated cuff inflation. In this study, a BPT-based prototype was developed and validated against a standard sphygmomanometer for BP estimation, while peripheral oxygen saturation (SpO₂), heart rate (HR), and skin temperature were concurrently measured and cross-validated with commercial devices. Recent research has explored cuffless blood pressure (BP) monitoring using photoplethysmography (PPG) and advanced computational methods. Joung et al. [6] proposed PPG2BP-Net, a 1D-CNN model trained on 4,185 subjects, which achieved mean errors of 0.209 ± 7.509 mmHg for systolic BP (SBP) and 0.150 ± 4.549 mmHg for diastolic BP (DBP), satisfying both AAMI and BHS standards. Cao et al. [7] introduced Crisp-BP, a wrist-worn device employing reflected wave transit time (RWTT), hybrid BLSTM networks, and transfer learning, which reported mean errors of 1.67 mmHg (SBP) and 0.86 mmHg (DBP) with standard deviations of 7.31 mmHg and 6.55 mmHg, respectively, also meeting AAMI/BHS criteria. Schrumpf et al. [8] applied CNN-based models (AlexNet, ResNet, and spectrotemporal networks) to both PPG and remote PPG signals, finding acceptable accuracy in normal BP ranges but significant errors at extreme values, falling short of AAMI/BHS requirements. Haddad et al. [9] proposed a multi-linear regression model utilizing 27 PPG morphological features and derivatives, achieving mean absolute errors of 6.10 mmHg (SBP) and 4.65 mmHg (DBP) with standard deviations of 8.08 and 6.22 mmHg, respectively, though requiring subject-specific calibration. Collectively, these studies highlight progress in continuous, cuffless BP estimation while underscoring limitations in generalization, robustness, and autonomous operation.

This research was conducted at *****.

II. METHODOLOGY

This section explains the methodology of the prototype's vital acquisition techniques that use BPT (blood pressure trend) to estimate blood pressure and measurement heart rate, oxygen saturation rate (SpO₂) and skin temperature.

A. Photoplethysmography (PPG) sensor

Cuffless estimation of blood pressure, peripheral oxygen saturation (SpO₂), and heart rate was performed using the MAX30102 optical sensor in conjunction with the MAX32664D biometric sensor hub. Both SpO₂ and blood pressure measurements required calibration. For SpO₂ assessment, three calibration coefficients were derived prior to subsequent signal processing. The computation of SpO₂ was carried out according to equations (1) and (2) [10]:

$$\text{SpO}_2 = a \cdot R^2 + b \cdot R + c \quad (1)$$

Where, sensor calibration values, a is 1.5958422, b is -34.659664, c is 112.68987 and R represents the PPG signal,

$$R = \frac{AC_{\text{red}}/DC_{\text{red}}}{AC_{\text{ired}}/DC_{\text{ired}}} \quad (2)$$

Where, AC and DC are pulsatile and non-pulsatile component respectively of Red and IR LED.

SpO₂ values are regarded as normal within the range of 95% to 100%. In a similar manner, three reference blood pressure measurements obtained using a sphygmomanometer are required to estimate cuffless blood pressure trends through the robotic arm. The calibration procedure typically requires approximately one minute to complete and can also be validated in patients receiving antihypertensive medication [11].

B. Skin Temperature Measurement Sensor

Body temperature varies depending on the site of measurement, with slight differences observed across different regions. Core temperature can be readily obtained using invasive methods, and values remain relatively consistent throughout the trunk and head. In contrast, skin temperature is highly influenced by the measurement site and ambient conditions [12]. Several pathological conditions can alter skin temperature, resulting in either elevated or reduced values. Since thermoregulation is a key component of homeostasis, such deviations may indicate underlying health problems. For example, insufficient red blood cell supply, as seen in anemia, can lead to cold extremities, while diabetic neuropathy may also cause cold hands [13]. In this study, skin temperature was measured at the palm using a medical-grade flexible cable integrated with the MAX30205 temperature sensor.

C. Block Diagram

The block diagram as shown in Fig. 1 of the proposed system illustrates a seamless integration of compact hardware modules for efficient physiological monitoring. At its core, the ESP8266 microcontroller orchestrates data acquisition and processing, interfacing with a MAX30102 PPG sensor, MAX32664 biometric hub [14], and MAX30205 temperature sensor [15] to capture real-time vital signs. Processed data is visually presented on a 1.8-inch RGB TFT LCD. Calibration data input is incorporated to maintain high measurement accuracy and adapt to individual user variability. The entire system is powered by a rechargeable battery setup, featuring a

TP4056 charging module, user-accessible switch, and indicator LED for reliable operation and power management.

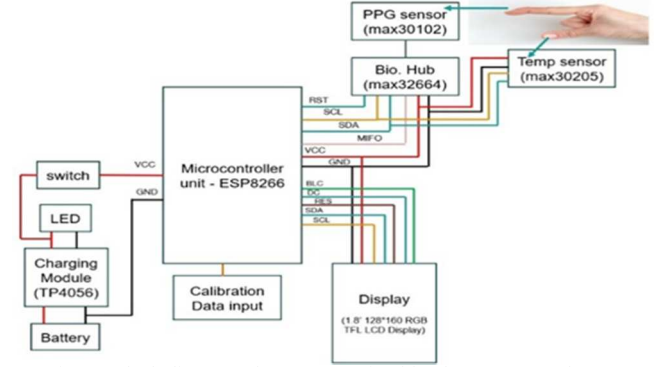


Fig. 1. Block diagram of prototype using blood pressure trend (BPT) technique.

D. Flow Chart

Fig. 2 outlines the system flow chart: power activation triggers the request for 2two adjacent fingers placement on the sensors. The PPG biometric sensor acquire SpO₂, heart rate, and BP trends; the temperature sensor on adjacent finger measures skin temperature on the finger. 3 times calibration data from a cuff device ensures accuracy which can last 1 week. The microcontroller signals, validates contact, and computes systolic/diastolic BP, heart rate, and SpO₂. Results are displayed on the TFT LCD and transmitted for remote monitoring.

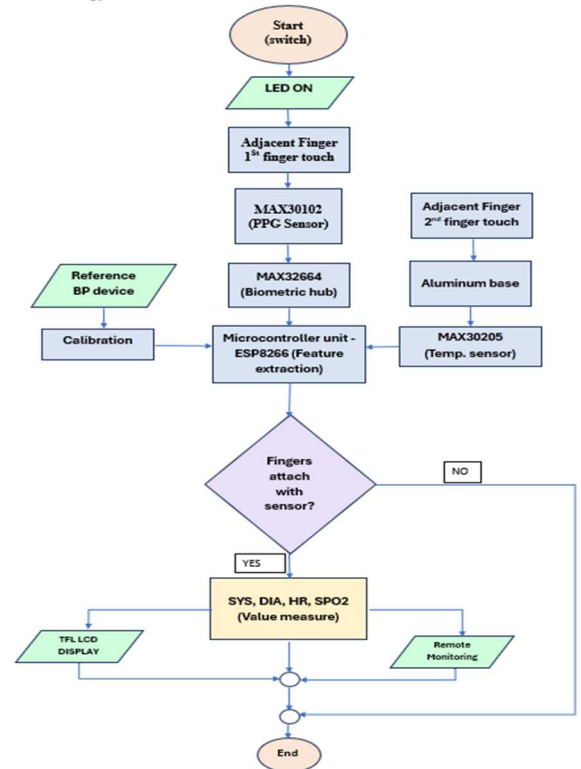


Fig. 2. Flow chart of prototype using blood pressure trend (BPT).

III. HARDWARE IMPLEMENTATION

The device was implemented based on block diagram that showed in figure 1. The diagram work as the system architecture for hardware implementation. The unit was powered by a Li-ion battery, ensuring portability and stable operation. A charging module with overcharge and discharge protection was added for safe recharging, along with an LED

indicator to operational status. An switch was placed between the battery and microcontroller. Together, the battery, charger, indicator, and switch ensured usability, safety, and protection against short circuits or overcurrent. At the core, the ESP8266 microcontroller powered by the regulated battery output acted as the system's brain. It processed and transmitted health data from two biomedical sensors connected: the Pulse Express Pulse-Ox & Heart Rate Sensor (MAX32664) and the Skin Temperature Sensor (MAX30205) as shown in Fig. 3.

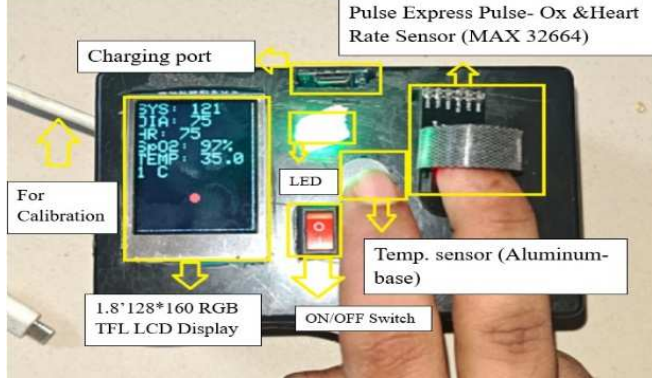


Fig. 3. Outer view of the prototype with PPG and temperature sensor reading.

When a finger was placed on the sensor area, real-time physiological data was captured with reduced noise due to short, secure wiring. Processed results were sent to a TFT LCD, displaying heart rate, SpO₂, estimated blood pressure, and temperature. The system was carefully designed with proper power distribution, grounding, and insulation to ensure both safety and reliable performance as shown in inside view of Fig. 4.

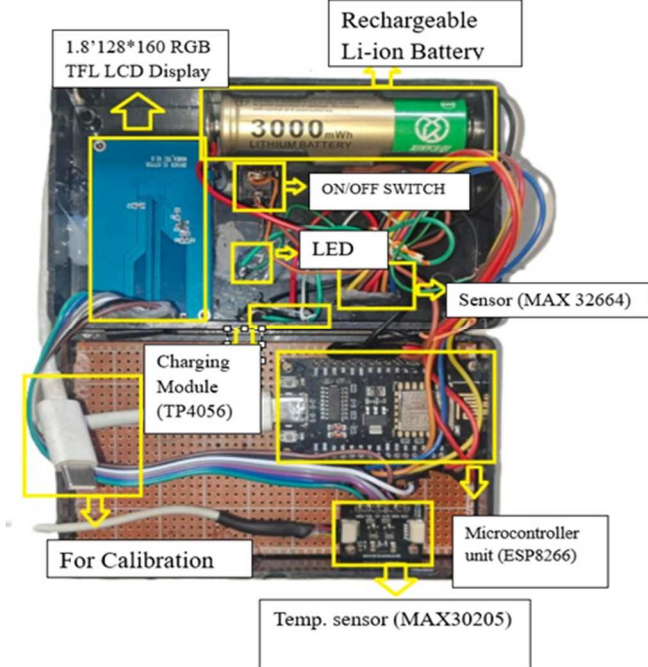


Fig. 4. Inside view of the designed prototype.

IV. RESULT ANALYSIS

The Institutional Review Board (IRB) of ***** following the Declaration of Helsinki, provided the ethical approval for the experiment. The test involved 10 volunteers in the 20 to 28 age range, equally divided between 5 male and 5 female students, as well

as 10 volunteers aged 45 to 64, including 5 males and 5 females. None of the participants reported any notable blood pressure illnesses.

For the Table I and Table II results, the all-over errors were calculated using Mean Absolute Percentage Error (3).

$$MAPE = \frac{1}{n} \sum_{t=1}^n \left| \frac{A_t - F_t}{A_t} \right| \quad (3)$$

Where, A_t is the actual value (acquired from cuff-based BP device), F_t is the forecast value (cuffless BP device) and n is the total number of observations.

The formula (4) for Mean Arterial pressure (MAP) is,

$$MAP = DBP + \frac{SBP - DBP}{3} \quad (4)$$

Where, SBP is systolic blood pressure and DBP is diastolic blood pressure.

Two sets of data were taken for accuracy measurement. Table I, consists of 10 volunteers (young age range 20-28) with data of blood pressure (systolic blood pressure and diastolic blood pressure), heart rate and oxygen saturation and MAP (mean arterial pressure) that measured by cuffless BPT. The MAPE percentage of everyone is also calculated in table. The MAPE percentage of each individual participant and for all the participants are also presented in the table. The MAPE% for systolic Blood Pressure is 3.47%, for diastolic blood pressure is 5.01%, for mean arterial pressure is 3.47%, for heart rate is 2.19% and for oxygen saturation is 0.83%. This error is comparatively lower. So, the accuracy for Systolic blood pressure is 96.53%, accuracy for diastolic blood pressure is 94.99%, accuracy for MAP is 96.53%, accuracy for heart rate is 97.52% and accuracy for oxygen saturation is 99.17% which is validated against pulse oximeter. The Table I. data is also plotted in Fig. 5, Fig. 6, Fig. 7 and Fig. 8 for blood pressure, MAP, heart rate and SpO₂ respectively.

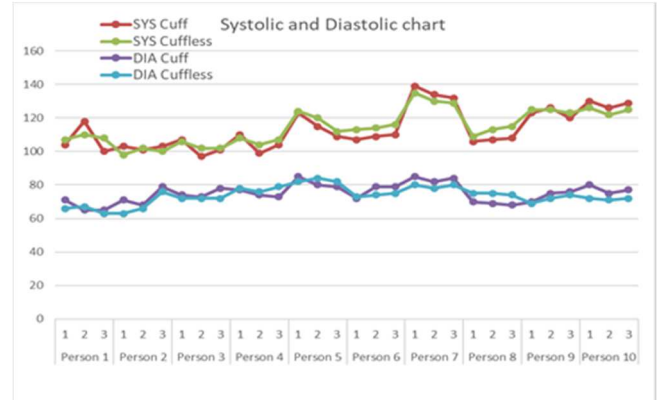


Fig. 5. Systolic and Diastolic data for cuff and cuffless (Age 20-28).

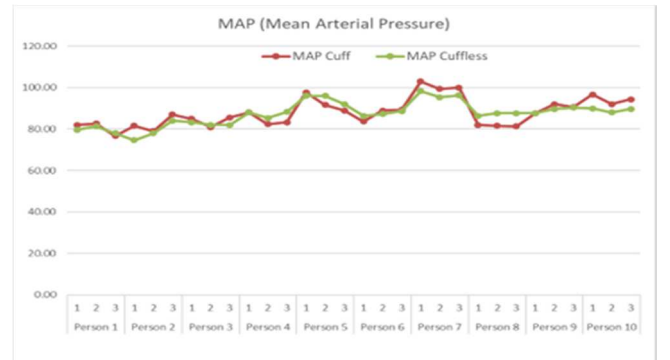


Fig. 6 Mean Arterial Pressure (MAP) data for cuff and cuffless (Age 20-28).

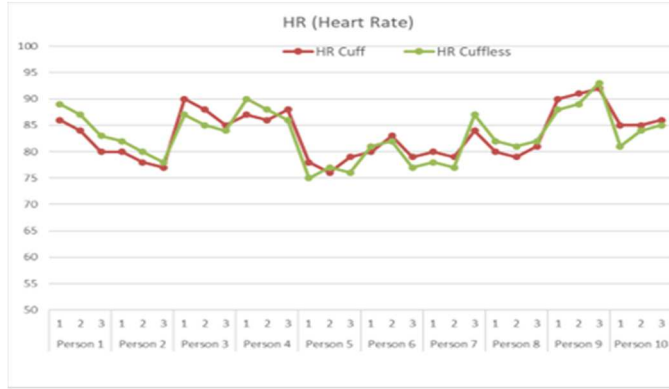


Fig. 6. Heart rate data for cuff and cuffless (Age 20-28).

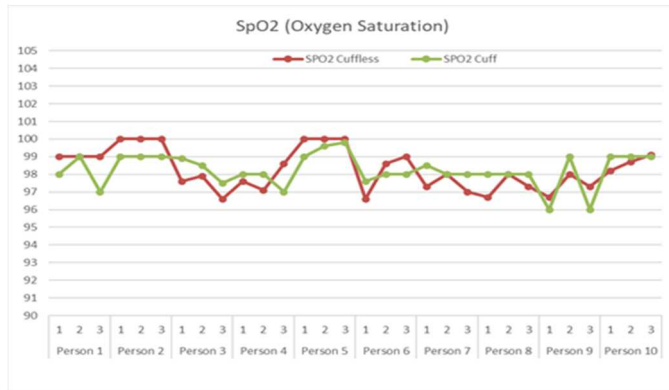


Fig. 7. Oxygen Saturation (SpO₂) data for pulse oximeter and cuffless (Age 20-28).

Table II, consist of 10 volunteers (elderly age range 45-64) with data of blood pressure (systolic blood pressure and diastolic blood pressure), heart rate and oxygen saturation and MAP (mean arterial pressure) that measured by cuffless BPT. The MAPE percentage of each individual participant and for all the participants are also presented on the table. The MAPE% for systolic Blood Pressure is 4.71%, for diastolic blood pressure is 3.47%, for mean arterial pressure is 3.47%, for heart rate is 2.19% and for oxygen saturation is 0.7%. So, the accuracy for Systolic blood pressure is 95.29%, accuracy for diastolic blood pressure is 94.91%, accuracy for MAP is %, accuracy for heart rate is 96.01% and accuracy for oxygen saturation is 99.30% which is validated against pulse oximeter. The Table I. data is also plotted in Fig. 8, Fig. 9, Fig. 10 and Fig. 11 for blood pressure, MAP, heart rate and SpO₂ respectively.

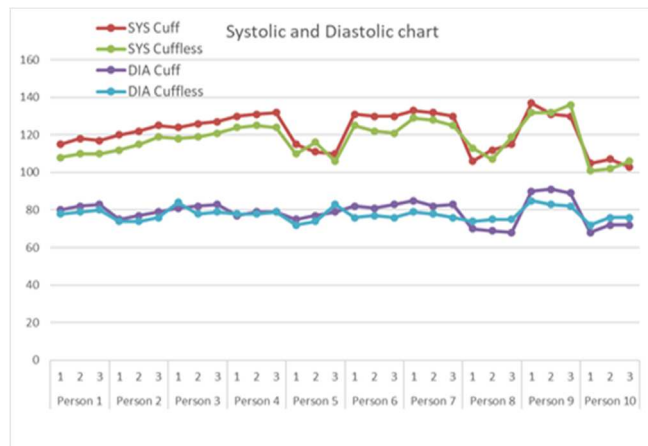


Fig. 8. Systolic and Diastolic data for cuff and cuffless (Age 45-64).

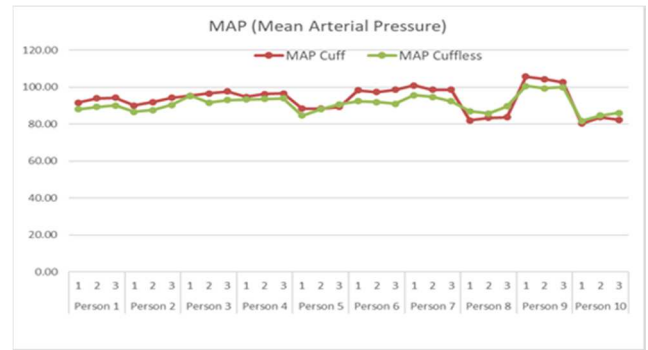


Fig. 9. Mean Arterial Pressure (MAP) data for cuff and cuffless (Age 45-64).

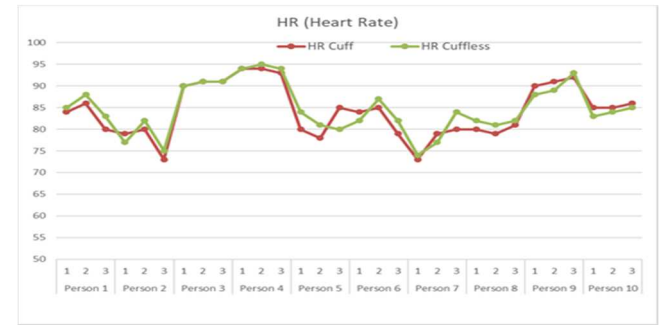


Fig. 10. Heart rate data for cuff and cuffless (Age 45-64).

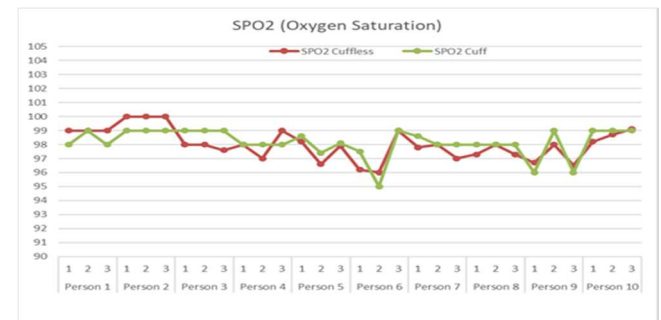


Fig. 11. Oxygen Saturation (SpO₂) data for pulse oximeter and cuffless (Age 45-64).

Individual measurements of blood pressure, heart rate, skin temperature, and oxygen saturation showed minor variations due to environmental and physiological factors such as stress, activity level, or time of day. Analysis of Table I and Table II revealed no significant accuracy difference between young and elderly participants, as neither group included subjects with hypertension or hypotension. For systolic blood pressure (SBP), a MAPE% of ~3.5% with an average reference of 120 mmHg resulted in an absolute error of ~4.2 mmHg, satisfying the AAMI requirement of ≤ 5 mmHg. For diastolic blood pressure (DBP), a MAPE% of ~5% with an average reference of 80 mmHg yielded an absolute error of ~4.0 mmHg, also within the 5 mmHg threshold. Heart rate and SpO₂ demonstrated even lower errors (2.2% and 0.8%, respectively), indicating high measurement precision. Moreover, the majority of SBP and DBP readings fell within 5–10 mmHg of the reference device, corresponding to BHS Grade A/B performance levels. Collectively, these findings confirm that the proposed cuffless system meets both AAMI and BHS clinical accuracy standards and provides reliable monitoring across different age groups. Additionally, the cost analysis presented in Table III demonstrates that the system can be manufactured at approximately 50 USD, making it an affordable and accessible solution for widespread healthcare implementation.

TABLE I. DATA TABLE FOR YOUNG PEOPLE AGE 20-28.

Subject	Data-Taken	Gender	Age	WEIGHT (KG)	Height (CM)	BMI	SYS Cuff	SYS Cuffless	DIA Cuff	DIA Cuffless	MAP Cuff	MAP Cuffless	MAPE SYS	MAPE DIA	MAPE SYS OVERALL	MAPE DIA OVERALL	MAPE MAP	MAPE MAP OVERALL	HR Cuff	HR Cuffless	HR MAPE	HR MAPE OVERALL	SPO2 Cuff	SPO2 Cuffless	SPO2 MAPE	SPO2 MAPE OVERALL																																													
Person 1	1	F	22	68	167.64	24.19	104	107	71	66	82.00	79.67	5.89	4.40	3.47	5.01	2.07	3.47	86	89	3.60	2.48	98	99	1.03	0.83																																													
	2						118	110	65	67	82.67	81.33							84	87			99	99																																															
	3						100	108	65	63	76.67	78.00							80	83			97	99																																															
Person 2	1	F	21	50	154.94	20.82	103	98	71	63	81.67	74.67	2.92	6.00			3.47		5.01	4.43	3.47		80	82	2.12		2.48	99	100	1.01	0.83																																								
	2						101	102	68	66	79.00	78.00											78	80				99	100																																										
	3						103	100	79	76	87.00	84.00											77	78				99	100																																										
Person 3	1	F	21	80	170.18	27.62	107	106	74	72	85.00	83.33	2.36	3.92						3.47			5.01	2.49	3.47			90	87	2.64		2.48	98.9	97.6	0.95	0.83																																			
	2						97	102	73	72	81.00	82.00																88	85				98.5	97.9																																					
	3						101	102	78	72	85.67	82.00																85	84				97.5	96.6																																					
Person 4	1	F	21	50	162.56	18.92	110	108	77	78	88.00	88.00	3.25	4.07										3.47				5.01	3.21	3.47			87	90	2.68		2.48	98	97.6	0.99	0.83																														
	2						99	104	74	76	82.33	85.33																					86	88				98	97.1																																
	3						104	107	73	79	83.33	88.33																					88	86				97	98.6																																
Person 5	1	F	22	47	160.02	18.35	123	124	85	82	97.67	96.00	2.64	4.11															3.47				5.01	3.27	3.47			78	75	2.99		2.48	99	100	0.54	0.83																									
	2						115	120	80	84	91.67	96.00																										76	77				99.6	100																											
	3						109	112	79	82	89.00	92.00																										79	76				99.8	100																											
Person 6	1	M	21	78	177.8	24.67	107	113	72	73	83.67	86.33	5.22	4.26																				3.47				5.01	1.94	3.47			80	81	1.66		2.48	97.6	96.6	0.89	0.83																				
	2						109	114	79	74	89.00	87.33																															83	82				98	98.6																						
	3						110	116	79	75	89.33	88.67																															79	77				98	99																						
Person 7	1	M	20	70	170.18	24.17	139	135	85	80	103.00	98.33	2.71	5.17																									3.47				5.01	4.07	3.47			80	78	2.87		2.48	98.5	97.3	0.75	0.83															
	2						134	130	82	78	99.33	95.33																																				79	77				98	98																	
	3						132	129	84	80	100.00	96.33																																				84	87				98	97																	
Person 8	1	M	28	50	167.64	17.79	106	109	70	75	82.00	86.33	4.97	8.22																														3.47				5.01	6.81	3.47			80	82	2.09		2.48	98	96.7	0.68	0.83										
	2						107	113	69	75	81.67	87.67																																									79	81				98	98												
	3						108	115	68	74	81.33	87.67																																									81	82				98	97.3												
Person 9	1	M	24	88	172.72	29.49	123	125	70	69	87.67	87.67	1.64	2.69																																			3.47				5.01	0.97	3.47			90	88	1.84		2.48	96	96.7	1.03	0.83					
	2						126	125	75	72	92.00	89.67																																														91	89				99	98							
	3						120	123	76	74	90.67	90.33																																														92	93				96	97.3							
Person 10	1	M	26	84	172.72	28.15	130	126	80	72	96.67	90.00	3.12	7.28																																								3.47				5.01	5.40	3.47			85	81	2.35		2.48	99	98.2	0.40	0.83
	2						126	122	75	71	92.00	88.00																																																			85	84				99	98.7		
	3						129	125	77	72	94.33	89.67																																																			86	85				99	99.1		

TABLE II. DATA TABLE FOR ELDERLY PEOPLE AGE BETWEEN 45-64.

Subject	Data-Taken	Gender	Age	WEIGHT (KG)	Height (CM)	BMI	SYS Cuff	SYS Cuffless	DIA Cuff	DIA Cuffless	MAP Cuff	MAP Cuffless	MAPE SYS	MAPE DIA	MAPE SYS OVERALL	MAPE DIA OVERALL	MAPE MAP	MAPE MAP OVERALL	HR Cuff	HR Cuffless	HR MAPE	HR MAPE OVERALL	SPO2 Cuff	SPO2 Cuffless	SPO2 MAPE	SPO2 MAPE OVERALL
Person 1	1	M	53	75	168	26.57	115	108	80	78	91.67	88.00	6.28	3.26	4.71	5.09	4.52	3.99	84	85	2.42	2.19	98	99	0.68	0.70
	2						118	110	82	79	94.00	89.33							86	88			99	99		
	3						117	110	83	80	94.33	90.00							80	83			98	99		
Person 2	1	F	48	73	155	30.39	120	112	75	74	90.00	86.67	5.73	3.01			4.22		79	77	2.59		99	100	1.01	
	2						122	115	77	74	92.00	87.67							80	82			99	100		
	3						125	119	79	76	94.33	90.33							73	75			99	100		
Person 3	1	M	54	75	172	25.35	124	118	81	84	95.33	95.33	5.04	4.47			3.32		90	90	0.00		99	98	1.14	
	2						126	119	82	78	96.67	91.67							91	91			99	98		
	3						127	121	83	79	97.67	93.00							91	91			99	97.6		
Person 4	1	F	47	95	162	36.20	130	124	77	78	94.67	93.33	5.09	0.85			2.31		94	94	0.71		98	98	0.68	
	2						131	125	79	78	96.33	93.67							94	95			98	97		
	3						132	124	79	79	96.67	94.00							93	94			98	99		
Person 5	1	F	45	47	152.4	20.24	115	110	75	72	88.33	84.67	4.16	4.32			2.01		80	84	4.91		98.6	98.2	0.48	
	2						111	116	77	74	88.33	88.00							78	81			97.4	96.6		
	3						110	106	79	83	89.33	90.67							85	80			98.1	97.9		
Person 6	1	M	58	84	176.78	26.88	131	125	82	76	98.33	92.33	5.89	6.90	6.45	84	82	2.84	97.5	96.2	0.80					
	2						130	122	81	77	97.33	92.00				85	87		95	96						
	3						130	121	83	76	98.67	91.00				79	82		99	99						
Person 7	1	M	48	70	170.18	24.17	133	129	85	79	101.00	95.67	3.29	6.79	5.25	73	74	2.97	98.6	97.8	0.61					
	2						132	128	82	78	98.67	94.67				79	77		98	98						
	3						130	125	83	76	98.67	92.33				80	84		98	97						
Person 8	1	F	49	68	167.64	24.20	106	113	70	74	82.00	87.00	4.85	8.23	5.36	80	82	2.09	98	97.3	0.48					
	2						112	107	69	75	83.33	85.67				79	81		98	98						
	3						115	119	68	75	83.67	89.67				81	82		98	97.3						
Person 9	1	M	64	72	172.72	24.13	137	132	90	85	105.67	100.67	3.01	7.40	4.04	90	88	1.84	96	96.7	0.75					
	2						131	132	91	83	104.33	99.33				91	89		99	98						
	3						130	136	89	82	102.67	100.00				92	93		96	96.5						
Person 10	1	F	55	61	157.48	24.60	105	101	68	72	80.33	81.67	3.80	5.66	2.44	85	83	1.56	99	98.2	0.40					
	2						107	102	72	76	83.67	84.67				85	84		99	98.7						
	3						103	106	72	76	82.33	86.00				86	85		99	99.1						
													Accuracy	95.29	94.91		96.01		Accuracy	97.81		Accuracy	99.30			

V. CONCLUSION

Blood pressure monitoring is essential for cardiovascular health, yet traditional cuff-based methods impose discomfort, interrupt continuous assessment, and can cause pain, vascular damage, or infection risk in vulnerable patients. These issues are particularly acute in critical care environments such as ICUs and CCUs, where uninterrupted monitoring of heart rate, oxygen saturation, and blood pressure is vital. This research introduces a practical, cost-effective cuffless continuous blood pressure monitoring system priced at approximately 50 USD, employing Blood Pressure Trending (BPT) technology with initial calibration against a standard device. Clinical validation demonstrated that most systolic and diastolic readings fell within 5–10 mmHg of reference measurements, achieving BHS Grade A/B performance and satisfying AAMI accuracy standards. Across both young adult and elderly cohorts, the system maintained high reliability, providing continuous blood pressure, heart rate, and SpO₂ data with mean absolute percentage errors below 5%. By eliminating inflatable cuffs after calibration, the portable, battery-powered solution enhances patient comfort and supports real-time data transmission for remote monitoring. This innovation aligns with SDG 3: Good Health and Well-being by improving diagnostic precision, reducing patient discomfort, and facilitating preventive care. Ultimately, the system strengthens healthcare delivery, reduces hospitalization burdens, and elevates the quality of care for elderly and critically ill patients, benefiting both caregivers and those under continuous cardiovascular surveillance.

REFERENCES

- [1] A. K. M. M. Islam and A. A. S. Majumder, "Hypertension in Bangladesh: a review," *Indian Heart J.*, vol. 64, no. 3, pp. 319–323, May 2012, doi: 10.1016/S0019-4832(12)60096-0.
- [2] "Preeclampsia: MedlinePlus Medical Encyclopedia." Accessed: Sept. 11, 2025. [Online]. Available: <https://medlineplus.gov/ency/article/000898.htm>
- [3] S. S. Mousavi, M. A. Reyna, G. D. Clifford, and R. Sameni, "A Survey on Blood Pressure Measurement Technologies: Addressing Potential Sources of Bias," *Sensors*, vol. 24, no. 6, p. 1730, Jan. 2024, doi: 10.3390/s24061730.
- [4] "Continuous cuffless blood pressure monitoring based on PTT | IEEE Conference Publication | IEEE Xplore." Accessed: Sept. 29, 2025. [Online]. Available: <https://ieeexplore.ieee.org/document/5479013>
- [5] "How To Take Blood Pressure Measurements." Accessed: Sept. 29, 2025. [Online]. Available: <https://my.clevelandclinic.org/health/diagnostics/25068-blood-pressure-measurement>
- [6] J. Joung et al., "Continuous cuffless blood pressure monitoring using photoplethysmography-based PPG2BP-net for high intrasubject blood pressure variations," *Sci. Rep.*, vol. 13, no. 1, p. 8605, May 2023, doi: 10.1038/s41598-023-35492-y.
- [7] Y. Cao, H. Chen, F. Li, and Y. Wang, "Crisp-BP: continuous wrist PPG-based blood pressure measurement," in *Proceedings of the 27th Annual International Conference on Mobile Computing and Networking*, in *MobiCom '21*. New York, NY, USA: Association for Computing Machinery, Oct. 2021, pp. 378–391. doi: 10.1145/3447993.3483241.
- [8] F. Schrupf, P. Frenzel, C. Aust, G. Osterhoff, and M. Fuchs, "Assessment of deep learning based blood pressure prediction from PPG and rPPG signals," in *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, June 2021, pp. 3815–3825. doi: 10.1109/CVPRW53098.2021.00423.
- [9] S. Haddad, A. Boukhayma, and A. Caizzone, "Continuous PPG-Based Blood Pressure Monitoring Using Multi-Linear Regression," *IEEE J. Biomed. Health Inform.*, vol. 26, no. 5, pp. 2096–2105, May 2022, doi: 10.1109/JBHI.2021.3128229.
- [10] Guidelines for SPO₂ measurement. (2023). Analog Devices. <https://www.analog.com/en/technical-articles/guidelines-for-spo2measurement--maxim-integrated.html>
- [11] A. Niktash, "Measuring Blood Pressure, Heart Rate, and SpO₂ Using MAX32664D - A Quick Start Guide". Available: <https://www.analog.com/media/en/technical-documentation/userguides/measuring-blood-pressure-heart-rate-and-spo2-usingmax32664d.pdf>
- [12] R. Lenhardt and D. I. Sessler, "Estimation of Mean-body Temperature from Mean-skin and Core Temperature," *Anesthesiology*, vol. 105, no. 6, Art. no. 6, Dec. 2006. Available: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1752199/>
- [13] "Diabetic neuropathy," *Nature Reviews Disease Primers*, vol. 5, no. 1. Springer Science and Business Media LLC, Jun. 13, 2019, doi: 10.1038/s41572-019-0097-9.
- [14] "MAX32664 Datasheet and Product Info | Analog Devices| Datasheet and Product Info|Analog Devices." Accessed: Sept. 25, 2025. [Online]. Available: <https://www.analog.com/en/products/max32664.html>
- [15] "MAX30205 Datasheet and Product Info | Analog Devices| Datasheet and Product Info|Analog Devices." Accessed: Sept. 25, 2025. [Online]. Available: <https://www.analog.com/en/products/max30205.html>